

Quest ArUco Recording Session Guide

Data collection for filter tuning · quest-aruco-avatar · September 2026

Session overview

#	Recording (test type)	Duration	Answers	Priority
1	stationary — 6 viewpoints	~3 min	Startup thresholds, viewpoint bias envelope	Required
2	moving — 105 mm A→B→A	~2.5 min	Movement tracking, re-anchor behaviour	Required
3	compressions — real CPR	~2 min	What compressions do to the markers and fusion	Required
4	hiding — cover each marker	~2 min	Fusion robustness when markers disappear	Recommended
5	calibration — 6 viewpoints	~2 min	Marker-to-anchor offsets for the offline replay	Only if markers were re-mounted

Total: about 15 minutes of recording plus setup. Do all recordings in **one session**, without restarting or recentering the Quest in between.

Controller buttons (identical in every recording)

- **Right A** — start / stop the recording.
- **Left X** while **stopped** — select the test type (cycles: calibration → stationary → headmotion → moving → hiding → compressions).
- **Left X** while **recording** — advance to the next phase.
- The floating label in the headset always shows the state, test type and phase (e.g. **RECORDING - stationary, Phase 2/6: stationary_left_view**). Trust the label, never memory.

Before you start (once)

1. Markers mounted rigidly in the usual arrangement; all three detected (gizmos react when you look at them).
2. Verify the new build: press **Left X** while stopped — the test type must cycle through **six** names ending in **compressions**. If it does not, the old APK is still installed.
3. Good, even lighting. No direct glare on the markers.
4. Clear ~2 m of space around the mannequin so you can reach every viewpoint.
5. Have ready: an A4 sheet (folded to 105 mm), tape, a stiff piece of cardboard (for hiding markers), pen and paper for notes.

Rules for every recording: if you press a wrong button, stop and redo the whole recording — never “fix” labels mentally. The mannequin must never be touched except where a step says to move it. Phase changes (Left X) must be pressed *exactly* when the described action starts or stops.

Recording 1 — stationary (6 viewpoints)

The mannequin never moves. Only you move. Distances are measured to the common/ID0 marker. Hold each viewpoint steady for **30 seconds**, keeping all markers in view.

1. While stopped, press **Left X** until the label shows **stationary**.
2. Stand **1.0 m directly in front** of the mannequin. Press **Right A**. Phase: **stationary_front**. Hold 30 s.
3. **Left X** → move to **45° to the left**, still ~1.0 m away. Hold 30 s. (**stationary_left_view**)
4. **Left X** → **45° to the right**, ~1.0 m. Hold 30 s. (**stationary_right_view**)
5. **Left X** → back to the front, headset **~30 cm higher** (stand tall / tiptoes). Hold 30 s. (**stationary_high_view**)
6. **Left X** → front, headset **~30 cm lower** (crouch). Hold 30 s. (**stationary_low_view**)
7. **Left X** → normal height, step back to **~1.5 m**. Hold 30 s. (**stationary_far_view**)
8. Press **Right A** to stop.

Recording 2 — moving (105 mm A→B→A)

Prepare the A and B marks

1. Pick one rigid, repeatable point on the mannequin close to the floor (a base corner, or tape a pointer). It must move rigidly with the markers.
2. Fold the 210 mm A4 width in half → a 105 mm ruler.
3. Tape mark **A** at the chosen point's current position; tape mark **B** exactly **105 mm horizontally** from A (towards your left when facing the mannequin).

wearer's left B <----- 105 mm -----> A (start)

Sequence — keep your head roughly still throughout

1. While stopped, select **moving** with **Left X**.
2. Base point on A. Press **Right A**. (**stationary_A**) Hold 30 s.
3. Press **Left X** *exactly as movement begins*. Slide the whole mannequin **rigidly** (no rotating, tilting or lifting) from A to B, spread evenly over **~30 s**. (**moving_A_to_B**)
4. Base point reaches B → press **Left X**. (**stationary_B**) Hold 30 s.
5. Press **Left X** as the return starts. Slide back over ~30 s. (**moving_B_to_A**)
6. Base point back on A → press **Left X**. (**stationary_A_return**) Hold 30 s.
7. Press **Right A** to stop.

Recording 3 — compressions (real CPR)

The mannequin stays in place. You kneel beside it in the genuine compression position. This recording deliberately captures occlusion by your hands and arms, markers flickering in and out, and the mannequin rocking — **do not try to keep the markers visible**.

1. While stopped, select **compressions** with **Left X**.
2. Kneel at the mannequin's side, hands OFF the chest, looking down at it as you would during CPR. Press **Right A**. (*still_before*) Hold still 30 s.
3. Press **Left X** *exactly as you begin*. Perform continuous chest compressions at a realistic **100–120 per minute** for **~60 s. (compressions)**
4. Press **Left X** *exactly when you stop*. Hands off, hold still 30 s. (*still_after*)
5. Press **Right A** to stop.

If the compressions shifted the mannequin, note roughly how far and in which direction — that drift is itself useful data. Do not push it back before the recording has stopped.

Recording 4 — hiding (cover each marker in turn)

The mannequin and your head stay still (stand ~1 m in front). Cover markers with a stiff piece of **cardboard held a few centimetres above** the marker — never touch the marker or the mannequin. Cover the whole marker including its white border; the other two must stay fully visible. Each phase lasts **~15 seconds**; press **Left X** at every change.

1. While stopped, select **hiding**. Press **Right A**. (*all_visible*) 15 s, nothing covered.
2. **Left X** → cover the **common / ID0** marker. 15 s. (*common_hidden*)
3. **Left X** → uncover. 15 s. (*all_visible*)
4. **Left X** → cover the **chest / ID1** marker. 15 s. (*chest_hidden*)
5. **Left X** → uncover. 15 s. (*all_visible*)
6. **Left X** → cover the **navel / ID2** marker. 15 s. (*navel_hidden*)
7. **Left X** → uncover. 15 s. (*all_visible*)
8. Press **Right A** to stop.

Recording 5 — calibration (only if markers were re-mounted)

Skip this if the marker arrangement is unchanged since the existing `data/calibration.csv` — offsets are marker-local and do not expire. If any marker was moved or re-glued, record: select **calibration**, keep **all three markers visible at all times**, mannequin untouched, and hold each phase ~20 s: **near** (~0.5 m) → **far** (~1.5 m) → **left_side** → **right_side** → **crouch** → **stand_tall**, pressing **Left X** between phases.

After the session

Write down (per recording)

- Order of takes, and which was a re-do.
- Moving: confirmed distance 105 mm, which base point, did it return exactly to A.
- Compressions: approximate compression rate, whether the mannequin drifted.
- Any accidental occlusion, tracking loss, wrong button press or interruption.

Copy the files to the PC

The CSVs are stored on the headset. With the Quest connected by USB:

```
adb pull /sdcard/Android/data/de.unigreifswald.opencvaruco/files/ data/session-2026-09/
```

Every file is named `aruco_<testtype>_<timestamp>.csv`, so takes cannot overwrite each other. Then the offline analysis (startup grid, compression inspection, estimator comparison) runs entirely from these files — no further headset time needed.

Generated from the project state of 3 September 2026 — logger version with six test types (aruco_csv_logger.gd). Companion file: docs/QUEST_FILTER_RECORDING_CHECKLIST.md.